Problem 1 (e):

From previous metrics, the smallest model was the int8 model, and multiple of the models performed very well, achieving an accuracy of 0.98.
By combining some of these model compression techniques, we can attempt to achieve a smaller size without sacrificing any accuracy. This is what I implemented.
After reviewing the results of size and accuracy on multiple different combinations of model compression techniques, I picked the smallest model that still had a very high accuracy.
I succeeded in reducing the size further by combining knowledge distillation with int8 quantization. By applying int8 quantization to the student model, I achieved a much smaller size of 3.62 KB compared to the original 6.10 KB of the unquantized student model without sacrificing accuracy.
The biggest change was using int8 quantization, which makes sense. In above experiments, the int8 model was always the smallest one, so applying it to the student knowledge distillation model here creates the smallest size model without reducing the performance.

Problem 2:

1) Edge Impulse free version has these types of learning blocks:
    a) Supervised learning: Classification
        i) Example: neural network (ex our iris classification lab)
    b) Supervised learning: Regression
        i) Example: neural network
    c) Unsupervised learning: anomaly detection
        i) Examples: GMM, k-means
2) These are the layers we have access to:
    a) Dense layer: fully connected layer to learn patterns
    b) 1D Convolution/pooling: applies filters to extract features
    c) 2D Convolution/pooling: applies filters to extract features
    d) Reshape: reshapes the data (ex 1D to 2D)
    e) Flatten: reshapes the data to 1D
    f) Dropout: randomly drops out some neurons to prevent overfitting
3) Under the Deployment section, the only model compression options are int8 quantization or unoptimized float32 model. However, the EON compiler can also optimize the model.
4) Methods for synthetic data generation:
    a) Black Forest Labs Flux-Pro synthetic image generator: Uses an AI model to generate images
    b) DALL-E 3 synthetic image generator: Uses OpenAI’s model to generate images
    c) Whisper synthetic voice generator: uses OpenAI’s text to speech model to generate synthetic voice samples

    Synthetic data is important in machine learning applications because it allows you to create data that might be hard to get in real life, and it is probably also cheaper and more abundant.